

Code Division for Multiple Access (CDMA): Basic CDMA Mechanism, Fundamentals of CDMA codes, Spreading Codes based on PN sequences, Correlation Properties of Random CDMA Spreading Sequences, Advantages of CDMA.

Orthogonal Frequency Division Multiplexing (OFDM): Introduction, Motivation and Multicarrier basics, OFDM basics, OFDM Example, MIMO OFDM, OFDM Peak to Average Power ratio, SC-FDMA.

[Text1: 5.1 to 5.5, 5.7, 7.1, 7.2, 7.3, 7.5, 7.7, 7.8]

Questions

1. Explain CDMA Mechanism. 10 marks
 2. Explain the fundamental properties of CDMA. 10 marks
 3. Explain the properties of PN sequences. 10 marks
 4. Explain correlation properties of random CDMA spreading sequences. 10 marks
 5. Explain advantages of CDMA. 10 marks
 6. Explain MIMO OFDM. 10 marks
 7. Note on SC-FDMA. 10 marks
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1. Explain CDMA Mechanism. 10 marks

CDMA (Code Division Multiple Access) is a wireless technology that allows multiple users to share a common radio channel simultaneously. This solves the fundamental problem of limited wireless spectrum.

Multiple Access (MA) technology is essential for wireless networks because, unlike a wired connection, all devices must share the same public airwaves.

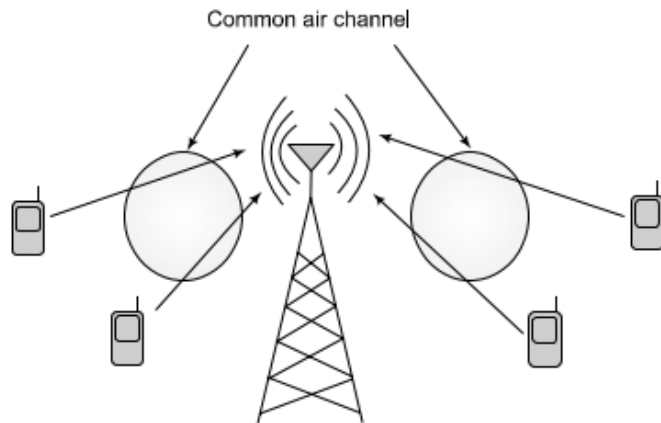


Figure 5.1 Multiple access for wireless cellular networks

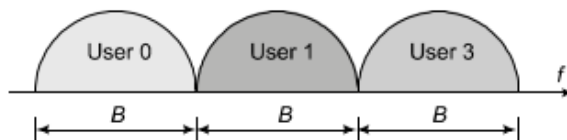


Figure 5.2 Frequency division for multiple access

1. Transmission (Combining Signals):

- Each user's data symbol (e.g., a_0 for user 0, a_1 for user1) is multiplied by a unique code sequence made of "chips" (e.g., $[1, 1, 1, 1]$ for User 0 and $[1, -1, 1, -1]$).
 - They encode it: $a_0 \times [1, 1, 1, 1] = [a_0, a_0, a_0, a_0]$
 - They encode it: $a_1 \times [1, -1, -1, 1] = [a_1, -a_1, -a_1, a_1]$
- This process "spreads" the user's symbol across multiple chips.
- The signals from all users are then added together into a single composite signal and transmitted.
 - Added $[(a_0+a_1), (a_0-a_1), (a_0-a_1), (a_0+a_1)]$

2. Reception (Separating Signals):

- To recover its data, a receiver correlates the composite signal with its own unique code.
 - They multiply the mixed signal by their code and add it up:
 - Multiply the added signal with secret code of user0
 - We get $(a_0+a_1) * 1 + (a_0-a_1) * 1 + (a_0-a_1) * 1 + (a_0+a_1) * 1 = 4a_0 \rightarrow$ data of user0
 - Multiply the added signal with secret code of i.e. $[1, -1, -1, 1]$
 - We get $(a_0+a_1) * 1 + (a_0-a_1) * -1 + (a_0-a_1) * 1 + (a_0+a_1) * -1 = 4a_1$ and User 0's message cancels out, leaving them with $4a_1$.
- This process works because the codes are orthogonal. When the composite signal is multiplied by the correct user's code and summed, the desired user's signal is reinforced, while the signals from other users cancel out.

Comparison with Other Multiple Access Technologies

- **FDMA (1G):** Separates users by assigning different **frequency** bands.
- **TDMA (2G):** Separates users by assigning different **time** slots.
- **CDMA (3G):** Separates all users by assigning different **codes**, allowing them to use the same frequency and time.

Working of CDMA

1. **Code Modulation:** Each user's data symbol is multiplied by a unique, assigned code (a sequence of "chips").
2. **Signal Combining:** The modulated signals from all users are added together into a single composite signal for transmission.
3. **Signal Recovery:** To retrieve a specific user's data, the receiver correlates the composite signal with that user's unique code. The orthogonal properties of the codes cause the desired user's signal to be reinforced while other users' signals cancel out.

Extra Information

Basic working of spread spectrum

Spread spectrum is a method of transmitting [radio signals](#) over a wide range of frequencies. It spreads the signal over a broader bandwidth than the minimum required to send the information, which provides advantages such as increased resistance to interference, improved security, and enhanced privacy.

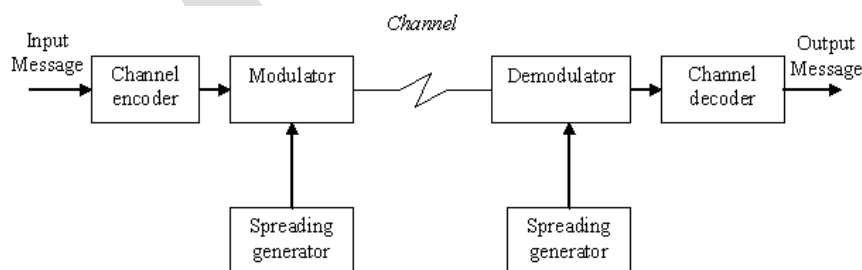


Figure 2: Block diagram of the spread spectrum communication system

1. Information Source

Provides the original message or data signal to be transmitted (e.g., binary data).

2. Spreading (Using PN Sequence)

The data signal is multiplied by a high-rate pseudo-noise (PN) sequence.

This spreads the signal bandwidth much wider than the original message band

3. Modulator

- The spread signal is **modulated** onto a carrier (e.g., using **BPSK or QPSK**).
- The wideband modulated signal is transmitted over the channel.

4. Channel

- The signal passes through a communication channel that may add **noise, fading, or interference**.

5. Demodulator

- At the receiver, the incoming wideband signal is demodulated to baseband.

6. Despreading (Using Same PN Sequence)

- The receiver multiplies the signal with the **same synchronized PN sequence**.
- This **compresses** the bandwidth back to the original narrowband signal.
- Unwanted signals or interference (not correlated with PN sequence) remain spread and appear as noise.

7. Output Data

- The despread signal is detected and converted back to the original information bits.

2. Explain Fundamental properties of CDMA codes. 10 marks

The reason CDMA relies "**Orthogonal Codes**." These codes have a mathematical property that allows them to cancel each other out.

1. The Key Property: Orthogonality

Orthogonality: CDMA uses orthogonal codes (like c_0, c_1, c_2, c_3) with zero correlation between them, allowing multiple users' signals to be separated from a composite signal.

Example

- Code for User 0: [1, 1, 1, 1]
- Code for User 1: [1, -1, -1, 1]
- Calculation: $(1*1) + (1*-1) + (1*-1) + (1*1) = 1 - 1 - 1 + 1 = 0$

This "zero" result is what causes **the interference from other users to cancel out** when you're trying to listen to your own signal.

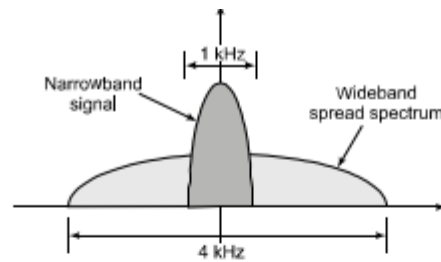


Figure 5.4 Spread spectrum communication

The different orthogonal spreading codes are

$$\begin{aligned} c_0 &= 1 \ 1 \ 1 \ 1 \\ c_1 &= 1 \ -1 \ -1 \ 1 \\ c_2 &= 1 \ -1 \ 1 \ -1 \\ c_3 &= 1 \ 1 \ -1 \ -1 \end{aligned}$$

The reader can verify that the codes c_0 , c_1 , c_2 , c_3 are orthogonal to each other. For example, consider c_1 , c_2 . The correlation r_{12} between codes c_1 , c_2 is

$$\begin{aligned} r_{12} &= \sum_{k=0}^3 c_1(k) c_2(k) \\ &= 1 \times 1 + (-1) \times (-1) + (-1) \times 1 + 1 \times (-1) \\ &= 1 + 1 + (-1) + (-1) \\ &= 0 \end{aligned}$$

2. Spreading & Bandwidth: In spread spectrum communication, each data symbol is multiplied by a special sequence called a spreading code (or PN code) of length N .

This process increases the bandwidth of the transmitted signal by a factor of N .

a) Original Signal

Suppose the data rate is 1 kbps (1000 bits per second).

This means each symbol lasts for

$$T = 1/1000 = 1 \text{ ms.}$$

The approximate bandwidth of this signal is

$$B = 1/T = 1 \text{ kHz.}$$

So, the original narrowband signal occupies about 1 kHz.

b) Spreading the Signal

Now, we use a spreading code of length $N = 4$.

Each data symbol is represented by 4 smaller bits, called chips.

To keep the same data rate (1 kbps), all 4 chips must be sent in the same 1 ms.

Therefore,

$$T_c = T/4 = 0.25 \text{ ms}$$

is the duration of each chip.

c) Increased Bandwidth

The signal bandwidth now depends on the chip rate, not the data rate.

The chip rate is

$$R_c = 1/T_c = 1/0.25 \text{ ms} = 4 \text{ kHz.}$$

Thus, the new bandwidth becomes 4 kHz, which is 4 times wider than the original 1 kHz.

3. System Capacity and Spectral Efficiency

a) Number of Users

If the spreading code length is N , then there can be at most N perfectly orthogonal codes.

Each user is assigned a unique spreading code.

Example:

For $N = 4$, there are 4 orthogonal codes (C_0, C_1, C_2, C_3).

Therefore, up to 4 users can transmit simultaneously without interfering with each other

b) Spectral Efficiency

At first, it may seem wasteful to increase the bandwidth by a factor of N

(e.g., from 1 kHz to 4 kHz).

However, this increase in bandwidth allows multiple users to share it at the same time.

	BW	Number of Users	Total Data Rate	Bandwidth per User
Original	1 kHz	1 user	1 kbps	1 kHz per user
CDMA	4 kHz	4 users	$4 \times 1 \text{ kbps} = 4 \text{ kbps}$	1 kHz per user

3. Explain the generation of PN sequences using LFSR and Properties of PN sequences 10 marks.

Spreading codes based on Pseudo-Noise (PN) sequences

The PN code is generated using a Linear Feedback Shift Register (LFSR) made up of several flip-flops (or delay elements) and mod-2 adders (XOR gates).

1. Components in the Diagram

Shift Registers (1, 2, ..., D):

Each block stores one bit and shifts its content to the next stage with every clock pulse.

For example, in your figure, there are D registers.

Feedback Taps:

The outputs from specific stages are XORed (added modulo-2) and fed back to the input of the first register.

Modulo-2 Adders ($\oplus$):

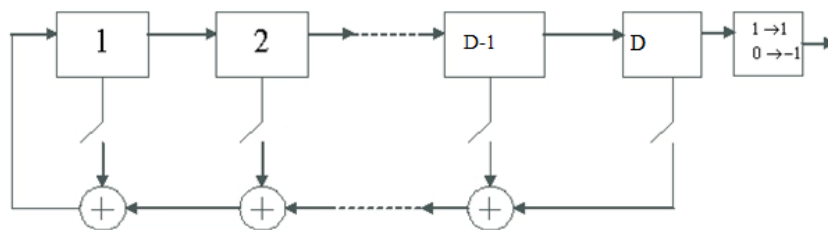
Perform binary addition without carry (i.e., $1 \oplus 1 = 0$).

Output Mapping:

Binary outputs are converted for signal modulation:

$1 \rightarrow +1$

$0 \rightarrow -1$



The primary method for generating these long, complex sequences is the **Linear Feedback Shift Register (LFSR)**.

Part-1 The Linear Feedback Shift Register (LFSR) Architecture

A Linear Feedback Shift Register (LFSR) is a circuit used to generate Pseudo Noise (PN) sequences.

It consists of a series of shift registers and XOR gates that produce a binary sequence which looks random but is completely deterministic.

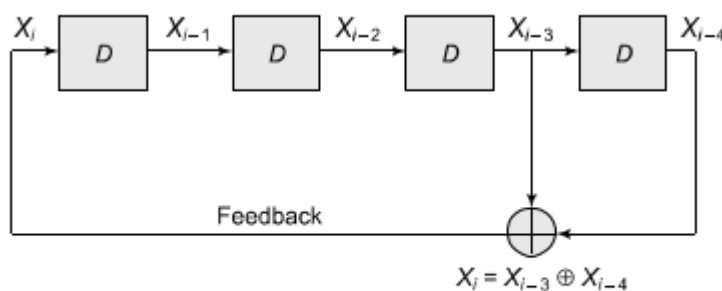


Figure 5.6 Linear feedback shift register

How It Works:

In the figure, there are 4 shift registers ($D = 4$).

At any instant, the register contents (or state) are: $X_{i-1}, X_{i-2}, X_{i-3}, X_{i-4}$

The feedback rule (governing equation) is:

$$X_i = X_{i-3} \oplus X_{i-4}$$

This means the new input bit is generated by taking the XOR of the third and fourth bits in the current state.

Step	Current	State New Bit (XOR)	New State (after shift)	Output Bit
1	1111	$1 \oplus 1 = 0$ 0111	1	
2	0111	$1 \oplus 1 = 0$ 0011	1	
3	0011	$0 \oplus 1 = 1$ 1001	1	
4	1001	$0 \oplus 1 = 1$ 1100	1	
5	1100	$1 \oplus 0 = 1$ 1110	0	
6	1110	$1 \oplus 0 = 1$ 1111	0	
7	1111	$1 \oplus 1 = 0$ 0111	1	
8	0111	$1 \oplus 1 = 0$ 0011	1	
9	0011	$0 \oplus 1 = 1$ 1001	1	
10	1001	$0 \oplus 1 = 1$ 1100	1	
11	1100	$1 \oplus 0 = 1$ 1110	0	
12	1110	$1 \oplus 0 = 1$ 1111	0	
13	1111	$1 \oplus 1 = 0$ 0111	1	
14	0111	$1 \oplus 1 = 0$ 0011	1	
15	0011	$0 \oplus 1 = 1$ 1001	1	

Possible States:

- A 4-bit shift register can have $2^4=16$ states (from 0000 to 1111).
- But an LFSR cannot use 0000, because if it becomes 0000, it will stay there forever (no new bits will be generated).

Maximum-Length

- So, it uses only 15 non-zero states $\rightarrow 2^4-1=15$
- When the LFSR cycles through all 15 states before repeating, it is called a Maximum-Length LFSR.
- The output sequence it generates is called a Maximum-Length PN (Pseudorandom Noise) sequence.

Example for 4-bit LFSR:

- Total output bits = 15 bits.
- Example PN sequence:
1 1 1 1 0 0 0 1 0 0 1 1 0 1 0

Mapping for CDMA (BPSK: Binary phase shift keying):

- In CDMA, bits are converted into symbols for spreading:
 - $0 \rightarrow +1$
 - $1 \rightarrow -1$
 - After mapping, PN sequence becomes:
-1 -1 -1 -1 +1 +1 +1 -1 +1 +1 -1 -1 +1 -1 +1
-

Properties of PN sequence

1. Balance Property

A PN (Pseudorandom Noise) sequence generated by a maximum-length LFSR has an almost equal number of +1s and -1s.

This property is called the Balance Property.

- For a shift register of length D , the total number of bits in the PN sequence is:

$$N = 2^D - 1$$

(It's always an **odd** number because the all-zero state is not used.)

- Since the total number of bits is odd, it's **impossible** to have exactly the same number of +1 and -1 values.
- However, the sequence is **almost balanced**:
 - Number of -1s (or 1s before mapping): 2^{D-1}
 - Number of +1s (or 0s before mapping): $2^{D-1} - 1$

Example (for $D = 4$)

- $N = 2^4 - 1 = 15$
- Number of -1s = $2^3 = 8$
- Number of +1s = $2^3 - 1 = 7$

So, in the PN sequence of 15 chips, there are **8 of one type** and **7 of the other** — almost equal.

2. Run-Length Property

A "run" is a string of consecutive identical chips. In a maximal-length PN sequence:

- **Half** of all runs are of length 1.
- **A quarter** of all runs are of length 2.
- **An eighth** of all runs are of length 3, and so on.
- The **total number of runs** is $2^{(D-1)}$.

This distribution of run lengths matches what is statistically expected from a random sequence, further reinforcing the noise-like quality of PN sequences.

3. Correlation Property (Most Important)

This is the most critical property for CDMA applications. It describes the autocorrelation function $r_{oo}(n_e)$ of the sequence with a version of itself shifted by n_e chips.

- **Zero Shift ($n_e = 0$):** The sequence perfectly aligns with itself, resulting in a correlation of **1**.
- **Any Non-Zero Shift ($n_e \neq 0$):** The correlation is **$-1/N$** , a very small value that approaches zero as the sequence length N increases.

This "thumbtack" autocorrelation profile—a sharp peak at zero shift and a low, constant value everywhere else—is crucial for:

- **Synchronization:** The receiver can find the correct timing by searching for the sharp correlation peak.
- **Signal Separation:** Signals from different users (or the same user with different time shifts) appear nearly uncorrelated, causing minimal interference.

4. Explain the Correlation properties of Random CDMA spreading sequences

The analysis assumes an ideal scenario where the spreading sequences are composed of Independent and Identically Distributed (IID) random chips. This means:

- Each chip $c_k(i)$ is either +1 or -1 with equal probability 1/2.
- The value of any chip is statistically independent of the value of any other chip, whether it's from the same user at a different time or from a different user.

This is modelled as:

$$P(c_k(i) = +1) = P(c_k(i) = -1) = \frac{1}{2}$$

$$E\{c_k(i)\} = 0$$

$$E\{c_k(i)c_l(j)\} = E\{c_k(i)\}E\{c_l(j)\} = 0 \quad (\text{for independent chips})$$

1. Autocorrelation Properties

The autocorrelation $r_{00}(k)$ measures the similarity between a user's code and a shifted version of itself.

- Zero Shift ($k=0$):

$$E\{r_{00}(0)\} = E\left\{\frac{1}{N} \sum_{i=0}^{N-1} c_0(i)c_0(i)\right\} = \frac{1}{N} \sum_{i=0}^{N-1} E\{c_0^2(i)\} = \frac{1}{N} \sum_{i=0}^{N-1} 1 = 1$$

This is a perfect "match," resulting in a correlation of 1.

- Non-Zero Shift ($k \neq 0$):

- Mean: $E\{r_{00}(k)\} = 0$. On average, the sequence is orthogonal to its shifted versions.
- Variance: $E\{r_{00}^2(k)\} = \frac{1}{N}$. While the average correlation is zero, for any single instance of a random sequence, the correlation at a non-zero shift will be a small, non-zero value. The power of this "correlation noise" is inversely proportional to the sequence length N .

Summary of Autocorrelation:

$$E\{r_{00}(k)\} = \begin{cases} 1 & \text{if } k = 0 \\ 0 & \text{if } k \neq 0 \end{cases}$$

$$\text{Variance} = \frac{1}{N} \quad (\text{for } k \neq 0)$$

2. Cross-Correlation Properties

The cross-correlation $r_{01}(k)$ measures the similarity between the codes of two different users (User 0 and User 1), with a relative shift k .

- Mean for any shift k : $E\{r_{01}(k)\} = 0$. On average, the codes of different users are orthogonal.
- Variance for any shift k : $E\{r_{01}^2(k)\} = \frac{1}{N}$. Again, for any specific pair of random sequences, the cross-correlation will be a small, non-zero value, with its power determined by $1/N$.

Key Implication and Comparison with Other Codes

The most important conclusion is that these random spreading codes are not perfectly orthogonal.

- Perfectly Orthogonal Codes (e.g., Walsh Codes): The cross-correlation is exactly zero for all shifts. This is ideal for separating users but requires perfect synchronization and a structured set of codes.
- Random Spreading Sequences: The cross-correlation has a mean of zero but a variance of $1/N$. This means that in practice, for any given set of user codes, there will be some small residual interference between users.

This residual interference is often called Multiple Access Interference (MAI). The performance of a CDMA system using random codes depends heavily on this variance. The longer the spreading sequence (the larger the NN), the smaller the interference, and the better the system performance.

5. Explain advantages of CDMA. 10 marks

a) Jammer margin

CDMA uses spread-spectrum technology. The jamming power is suppressed by a factor of N , which is the spreading factor (the length of the CDMA code). The SINR in a CDMA system becomes:

$$\text{SINR} = \frac{P}{\frac{P_j}{N} + \frac{\sigma_w^2}{N}}$$

The larger the spreading factor N , the greater the jammer suppression. This performance gain is called the **jammer margin**. Because of this robust feature, CDMA was originally developed for military use before being adopted in civilian cellular networks.

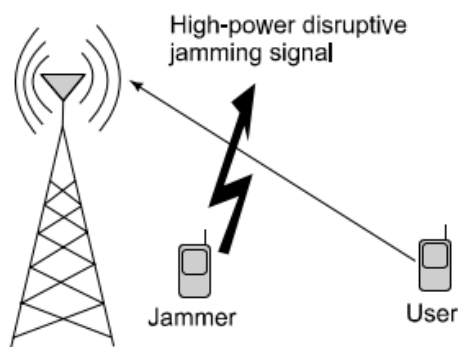


Figure 5.8 Disruption by jammer in wireless communication

b) Graceful degradation

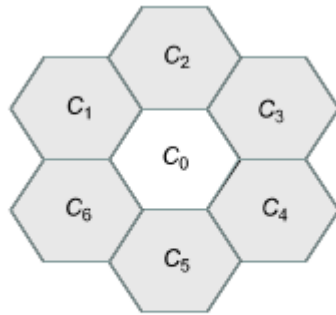
In a CDMA system, when a new user joins the network, the additional interference it creates is not concentrated on a single user. Instead, the interference is spread across all existing users.

The interference from any user is suppressed by the spreading factor N . Therefore, a new user ($K+1$) only adds a small, incremental amount of interference (specifically, $P_{\{K+1\}}/N$) to every other user in the system.

This "sharing" of the interference load means that no single user experiences a sudden, significant drop in performance. **Instead, the Signal-to-Interference-plus-Noise Ratio (SINR) for all users decreases gradually, or gracefully.**

This property is a key reason CDMA networks can implement universal frequency reuse (using the same frequency in every cell), leading to higher overall network capacity and spectral efficiency.

c) Frequency reuse



GSM: $K=7$, Seven cells use all the available frequencies and is called Reusefactor

Figure 5.10 Grid or lattice of hexagonal cells

GSM

In frequency reuse the **same radio frequencies** are used on **different transmitter towers (cell sites)** within a geographic area **that is separated by a sufficient distance to avoid interference**.

To prevent the same frequencies from interfering with each other, they are only reused in cells that are a safe distance apart. This is organized into a **Reuse Pattern or Cluster**.

- Cluster Size ($K=7$): A cluster is a group of cells that together use the complete set of available frequencies. No frequency is used twice within the same cluster. The number of cells in a cluster is the reuse factor.
- The 1/7th of the total available radio spectrum is available for use in any single cell.
- The total data rate and capacity of each cell is reduced to 1/7th of what it could be if the entire spectrum were used.

In traditional (GSM): Efficiency is proportional to $(1/7) \times \text{Total Bandwidth}$.

CDMA

In CDMA, all users transmit at the same time and on the same frequency. They are separated not by **frequency or time, but by unique codes**.

CDMA: Efficiency is proportional to $(1/1) \times \text{Total Bandwidth}$.

This is a theoretical 7x increase in spectral efficiency and network capacity compared to the GSM